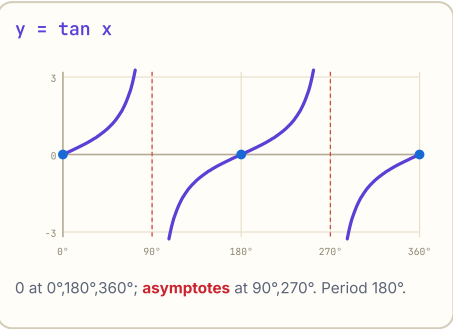
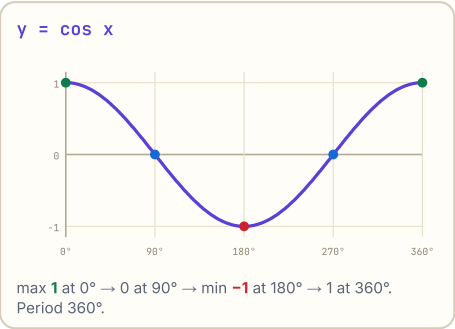
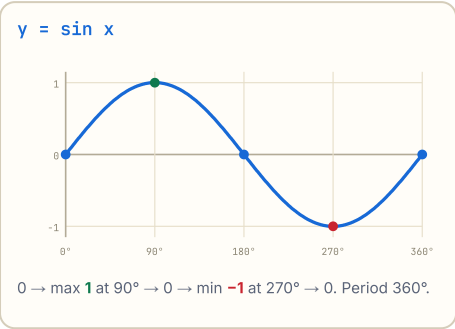


Trig Graphs & Equations — Cheat Sheet

Topic 6.3–6.4 · the three graphs over 0° – 360° , exact values, and the two-solution method.

round angles
to 1 d.p.

1 The three graphs · $0^\circ \leq x \leq 360^\circ$



2 Feature table

FEATURE	SIN X	COS X	TAN X
Start (0°)	0	1	0
Range	$-1..1$	$-1..1$	all $\mathbb{R}$
Period	360°	360°	180°
Max	1 @ 90°	1 @ $0, 360^\circ$	—
Min	-1 @ 270°	-1 @ 180°	—
Zeros	$0, 180, 360^\circ$	$90, 270^\circ$	$0, 180, 360^\circ$
Undefined	—	—	$90, 270^\circ$

3 Exact values (no calculator)

X	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	—

HAND TRICK

Under $0^\circ \cdot 30^\circ \cdot 45^\circ \cdot 60^\circ \cdot 90^\circ$ write **0 1 2 3 4**. sin = $\sqrt{n} / 2$; cos = same list reversed; tan = sin $\div$ cos.

Solving Trig Equations

Find the calculator angle, then use symmetry for its partner. Two answers in 0° – 360° .

$|k| > 1 \rightarrow$
no solution

THE PARTNER RULES · $0^\circ \leq x \leq 360^\circ$

Calculator gives first angle x_1 . Second solution:

- $\sin x = k \rightarrow 180^\circ - x_1$
- $\cos x = k \rightarrow 360^\circ - x_1$
- $\tan x = k \rightarrow 180^\circ + x_1$

Then $+360^\circ$ to any negative answer; keep only 0° – 360° .

METHOD (EVERY TIME)

- Rearrange to $\sin/\cos/\tan x = k$.
- First angle from the **positive** value of k .
- Apply the matching partner rule.
- Adjust into range; check both fit.

No solution if $|k| > 1$ for sin or cos. tan always gives two.

4 Worked examples

$$2 \cos x + 1 = 0$$

$$\begin{aligned}\cos x &= -\frac{1}{2} \\ \cos^{-1}\left(\frac{1}{2}\right) &= 60^\circ \\ -ve &\rightarrow 180 \pm 60 \\ x &= 120^\circ, 240^\circ\end{aligned}$$

$$\sin x = 0.6$$

$$\begin{aligned}x_1 &= \sin^{-1}0.6 = 36.9^\circ \\ 180 - 36.9 &= 143.1^\circ \\ x &= 36.9^\circ, 143.1^\circ\end{aligned}$$

$$\tan x = -1.5$$

$$\begin{aligned}\text{from } +1.5: & 56.3^\circ \\ 180 - 56.3, & 360 - 56.3 \\ x &= 123.7^\circ, 303.7^\circ\end{aligned}$$

WHY TWO ANSWERS

"Solve $\sin x = 0.5$ " = where does the flat line $y = 0.5$ cut the sine wave? It hits **twice** in 0° – 360° : the calculator gives one crossing (30°), the graph's symmetry gives the other (150°).

MEMORY HOOKS

Sine $\rightarrow 180 - x$ · Cos $\rightarrow 360 - x$ · Tan $\rightarrow 180 + x$
sin & cos **reflect**; tan **repeats every 180°** . sin/cos stuck in $-1..1$; tan reaches every value.

DIFFERENT RANGE?

If the interval isn't 0° – 360° (e.g. -180° to 180°), find all answers with the partner rule, then shift each by $\pm 360^\circ$ until it lands inside the range asked.